

RESOURCE MANAGEMENT



http://www.in.all.biz/img/in/service_catalog/22027.png

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Constraints on Projects

As described in the very beginning, all projects run in an environment of **constraints**.

1) TECHNICAL CONSTRAINTS

2) PHYSICAL CONSTRAINTS

3) RESOURCE CONSTRAINTS

- People
- Materials
- Space
- Money
- Equipment



<http://www.absecmy.com/project-theme/images/template/icons/resources.jpg>

In this lecture we look in detail at the **resource constraints** and techniques to improve their utilization through **Resource Leveling** and **Resource Smoothing**.



Resource Leveling

Defined by PMBOK as:

“A technique in which start and finish dates are adjusted based on resource constraints with the goal of balancing demand for resources with the available supply.”

Resource Leveling

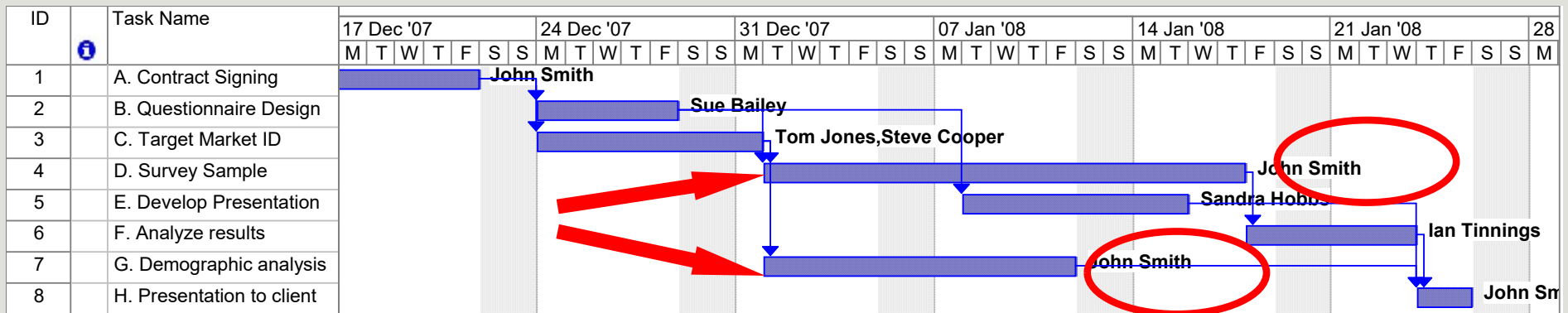
Assumption is that you are working in a "Project Organization", meaning everyone of the team is working 100% on your project.

Resource Leveling has two objectives:

1. To determine the **resource requirements** so that they will be available at the right time.
2. To allow each **activity** to be scheduled with the smoothest possible transition across resource usage levels.



Resource Loading Chart demonstrating Over-Allocation



With a **Gantt chart** like this it is easy to see that **John is over-allocated**. This assumes that he is utilized 100 % of the time every day. What if this was not the case.

Would you be able to tell?

If you create a Gantt chart that shows resources, you could determine if the scheduling above would work.

Resource Loading Chart demonstrating Over-Allocation

ID		Resource Name	Details	31 Dec '07							07 Jan '08							14 Jan '08	
				M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T
1	 	John Smith	Work		2d	2d	2d	2d			2d	2d	2d	2d	2d			1d	1d
		A. Contract Signing	Work																
		D. Survey Sample	Work		1d	1d	1d	1d			1d	1d	1d	1d	1d			1d	1d
		G. Demographic analy	Work		1d	1d	1d	1d			1d	1d	1d	1d	1d				
		H. Presentation to clie	Work																
2		Sue Bailey	Work																
		B. Questionnaire Desig	Work																
3		Tom Jones	Work	1d															
		C. Target Market ID	Work	1d															
4		Steve Cooper	Work	1d															
		C. Target Market ID	Work	1d															
5		Sandra Hobbs	Work									1d	1d	1d	1d			1d	1d
		E. Develop Presentati	Work									1d	1d	1d	1d			1d	1d
6		Ian Tinnings	Work																
		F. Analyze results	Work																

In this chart you group the resources and then list the tasks each has been assigned to. In this manner it is easy to see the amount of time that each resource is going to be utilized each day.

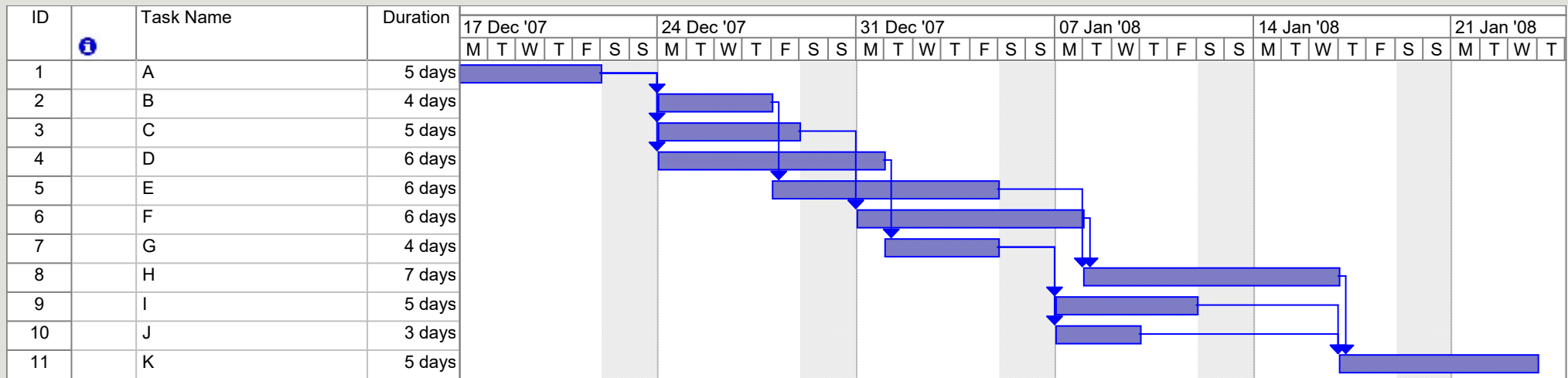
ID		Resource Name	Details	31 Dec '07							07 Jan '08							14 Jan '08	
				M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T
1	 	John Smith	Work		1d	2d	1d	2d			2d	2d	2d	1d	2d			1d	1d
		A. Contract Signing	Work																
		D. Survey Sample	Work		.5 d	1d	.75 d	1d			1d	1d	1d	.5 d	1d			1d	1d
		G. Demographic analy	Work		.5 d	1d	.25 d	1d			1d	1d	1d	.5 d	1d				
		H. Presentation to clie	Work																
2		Sue Bailey	Work																
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		E. Develop Presentati	Work									1d	1d	1d	1d			1d	1d
6		Ian Tinnings	Work																
		F. Analyze results	Work																

Rules of Thumb when solving Resource Leveling problems

When dealing with Resource Leveling problems deal with the tasks in this order:

1. Activities with the smallest amount of slack;
2. Activities with the smallest duration;
3. Activities with the lowest activity identification number (*e.g. those that start earliest in the WBS*);
4. Activities with the most successor tasks; and
5. Activities requiring the most resources.





	Activity	Duration	Predecessors	ES	EF	LS	LF	Slack
6								
2	A	5	-	0	5	0	5	0
4	B	4	A	5	9	6	10	1
3	C	5	A	5	10	5	10	0
3	D	6	A	5	11	8	14	3
3	E	6	B	9	15	10	16	1
2	F	6	C	10	16	10	16	0
4	G	4	D	11	15	14	18	3
4	H	7	E, F	16	23	16	23	0
3	I	5	G	15	20	18	23	3
4	J	3	G	15	18	20	23	5
2	K	5	H, I, J	23	28	23	28	0



Resource Loading Chart

Resource level limit: **9 people/day**

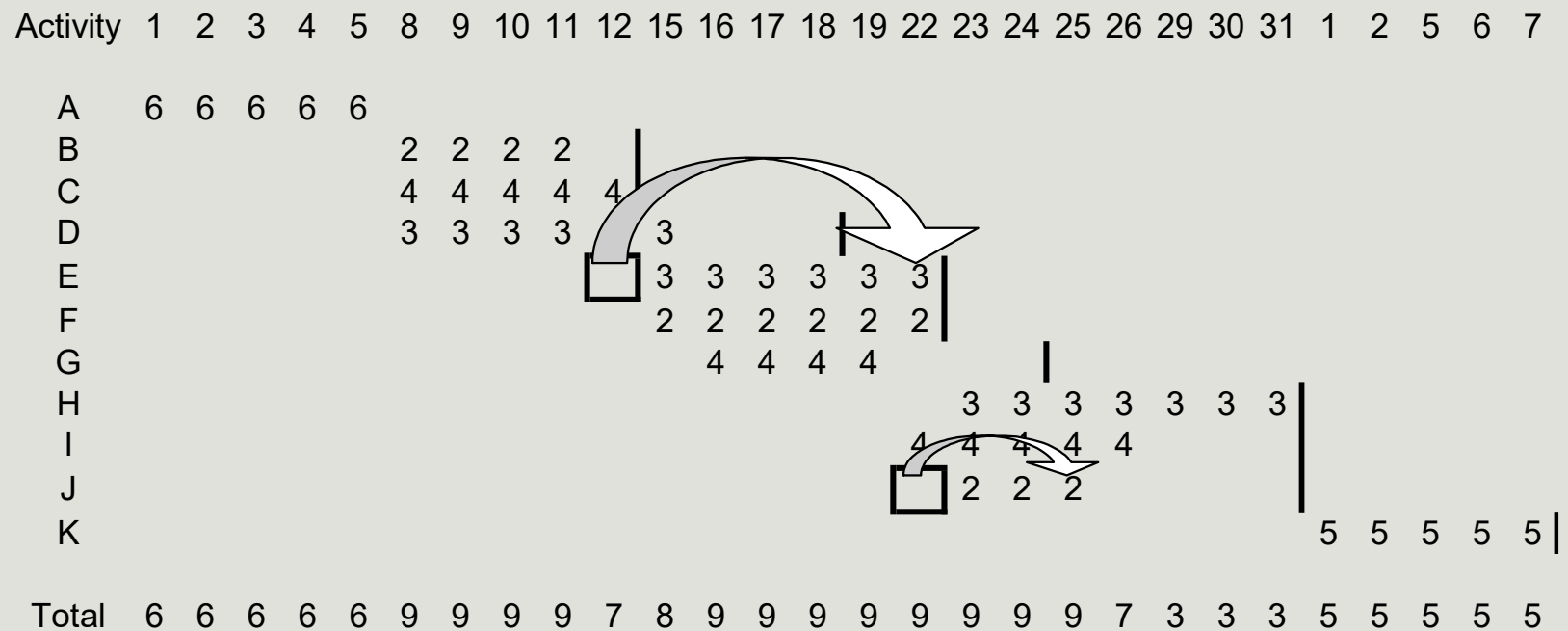
Before Resource Leveling

Activity	1	2	3	4	5	8	9	10	11	12	15	16	17	18	19	22	23	24	25	26	29	30	31	1	2	5	6	7
A	6	6	6	6	6																							
B						2	2	2	2																			
C						4	4	4	4	4																		
D						3	3	3	3	3	3																	
E										3	3	3	3	3	3													
F											2	2	2	2	2	2												
G												4	4	4	4													
H																	3	3	3	3	3	3	3					
I																	4	4	4	4	4							
J																	2	2	2									
K																								5	5	5	5	5
Total	6	6	6	6	6	9	9	9	9	10	8	9	9	9	9	8	9	9	7	7	3	3	3	5	5	5	5	5

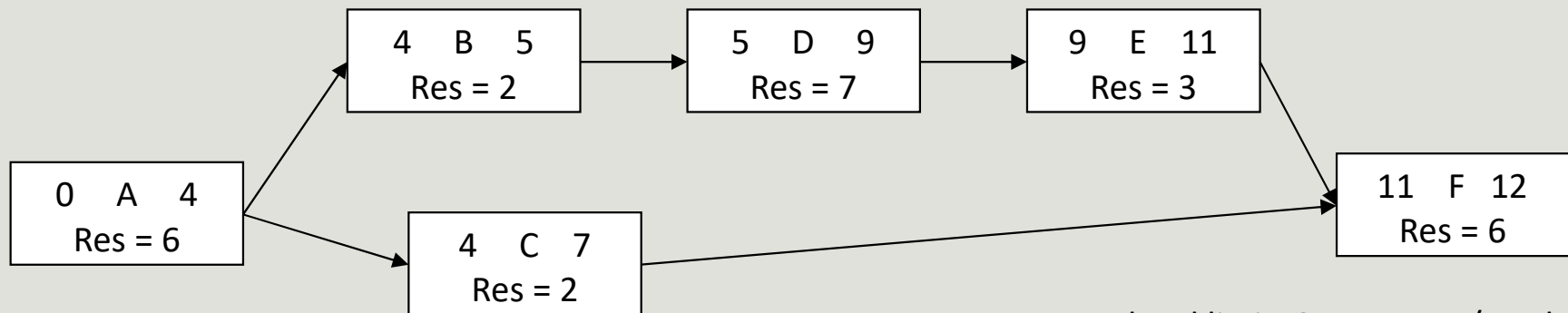


Resource level limit: 9 people/day

After Resource Leveling



Sample Project Network



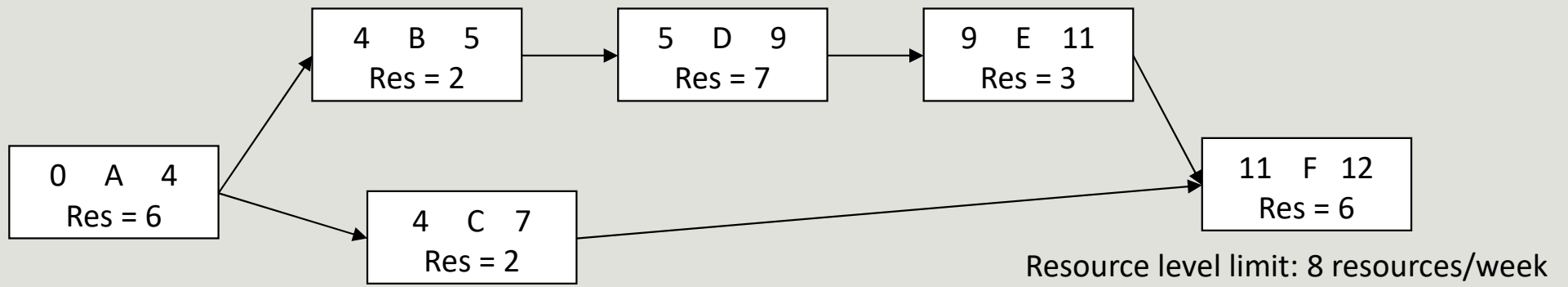
Resource level limit: 8 resources/week

Activity	Duration	Predecessors	ES	EF	LS	LF	Slack	Resource Hours Needed per Week	Total Resources Required
A	4	-	0	4	0	4	0	6	24
B	1	A	4	5	4	5	0	2	2
C	3	A	4	7	8	11	4	2	6
D	4	B	5	9	5	9	0	7	28
E	2	D	9	11	9	11	0	3	6
F	1	E,C	11	12	11	12	0	6	6
Total									72

Task C is the only one with slack available.

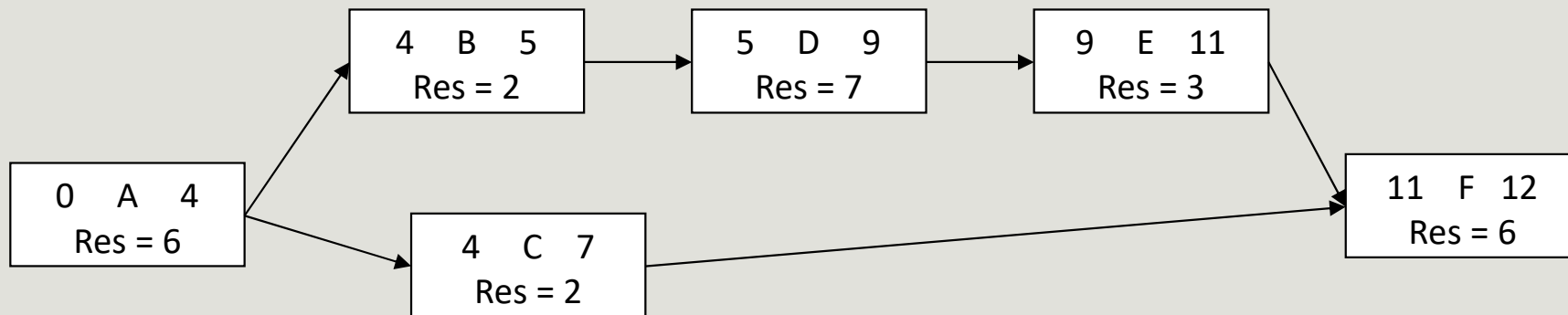
Remember first to level those tasks with the **smallest amount of slack**.

All tasks are on critical path, except C. Therefore, they all have priority over C

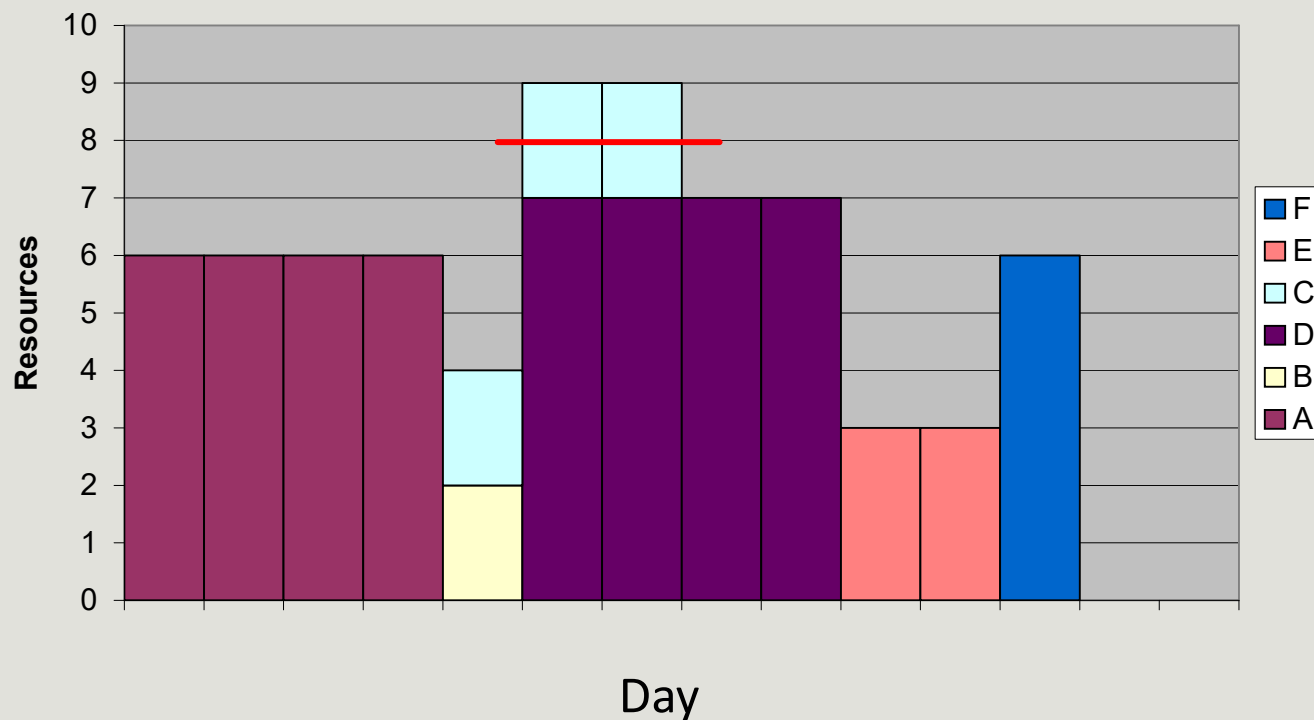


C is the only task available for leveling => find new home for it.





Resource Load Chart before leveling



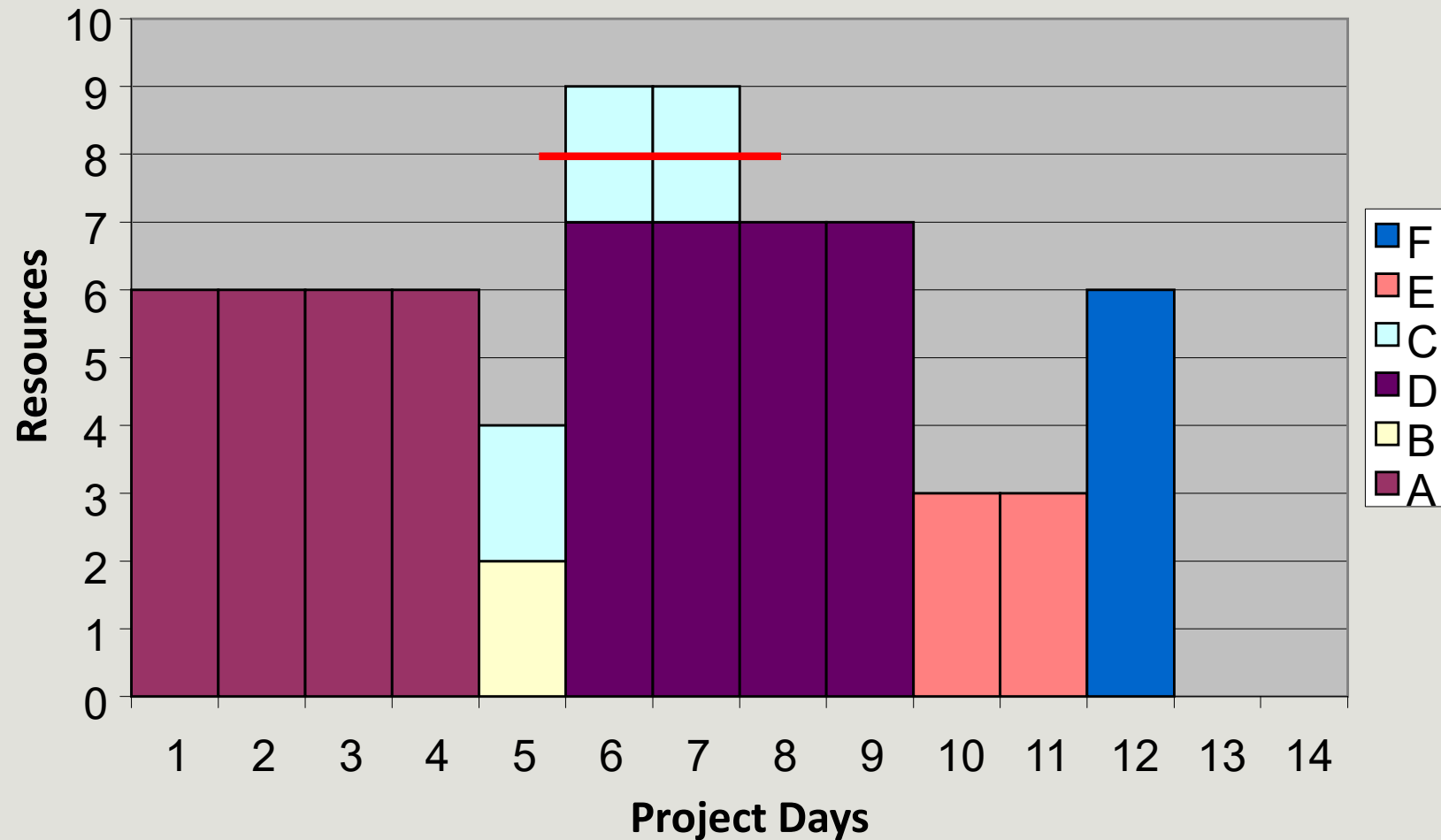
C is the only task available for leveling => find new home for it.



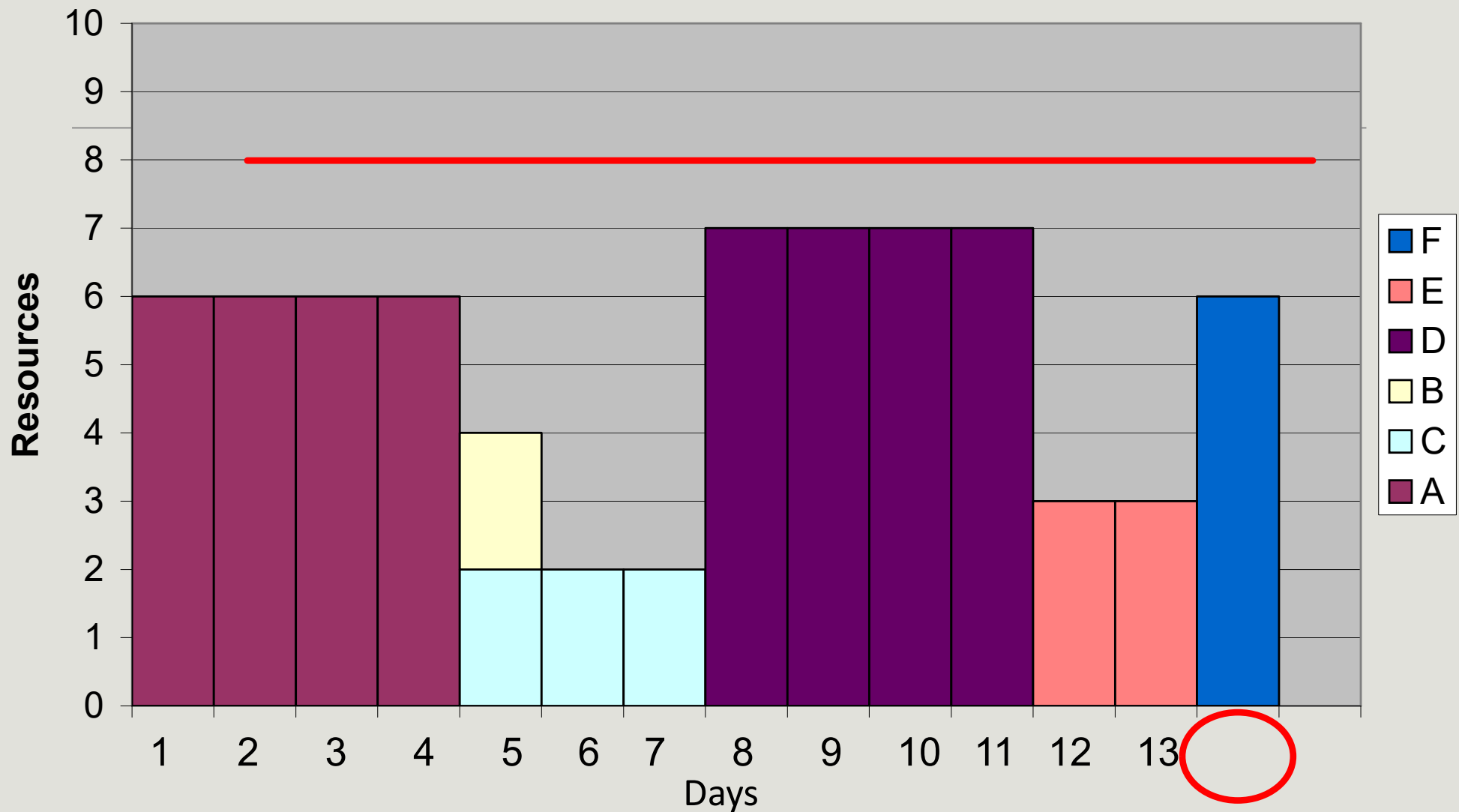
Resource Loading Charts

Resource loading charts used to display the amount of resources required as a function of time on a graph.

Each **activity** is typically represented in the charts as a block.

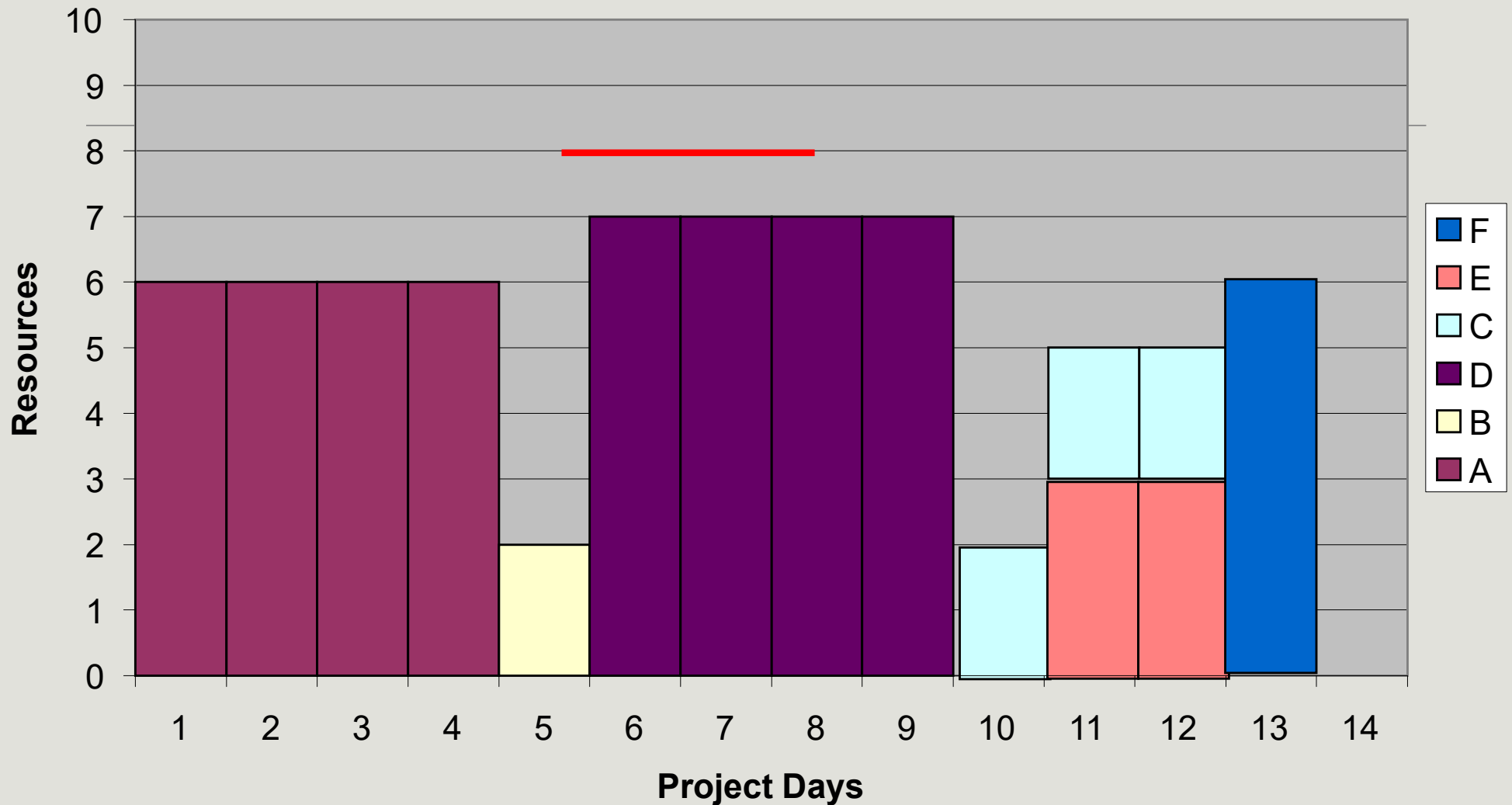


Resource Load Chart after leveling



How can we let the project finish on the original end day (12)?

Resource Load Chart before leveling



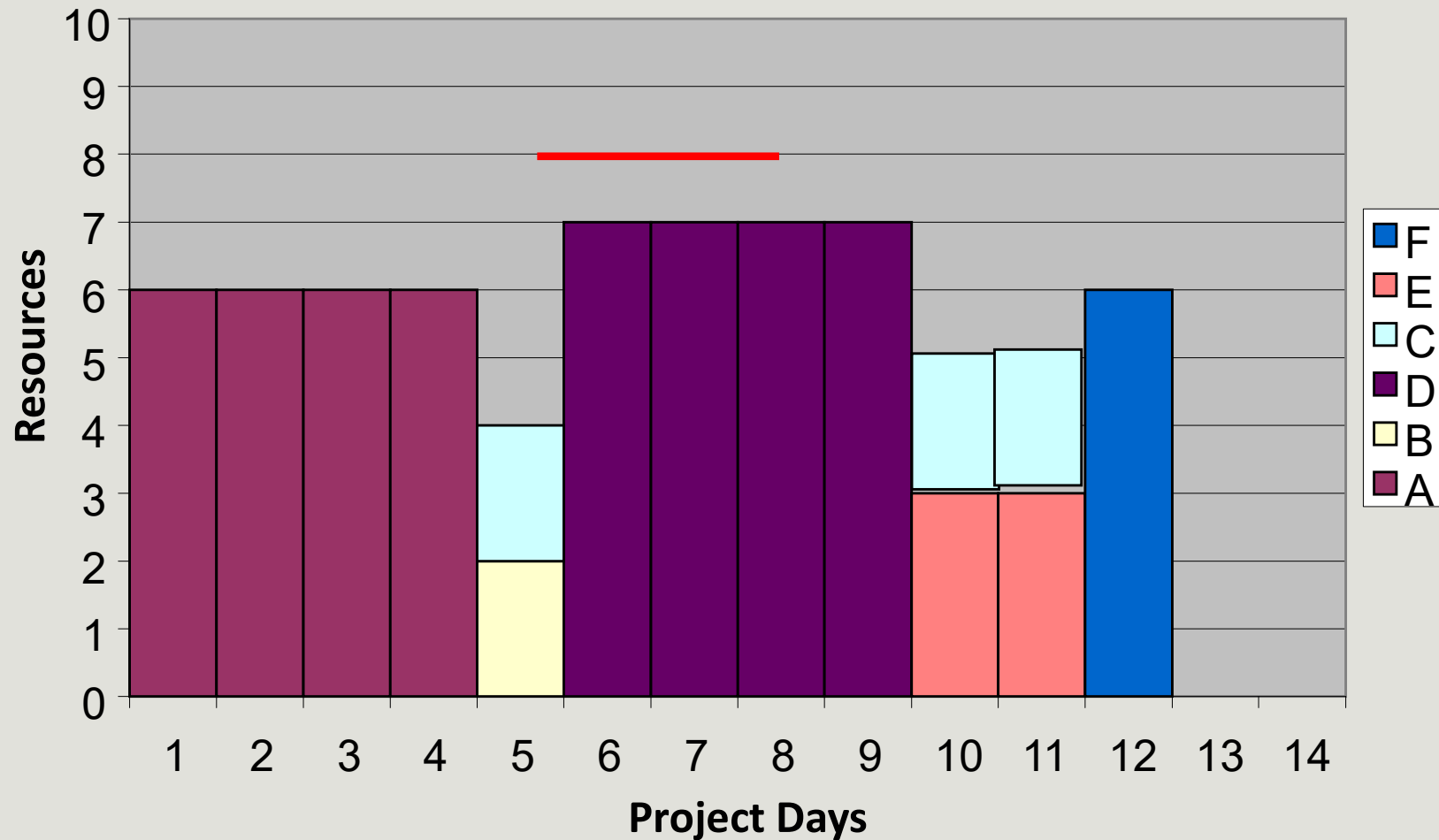
C is the only task available for leveling => find new home for it.



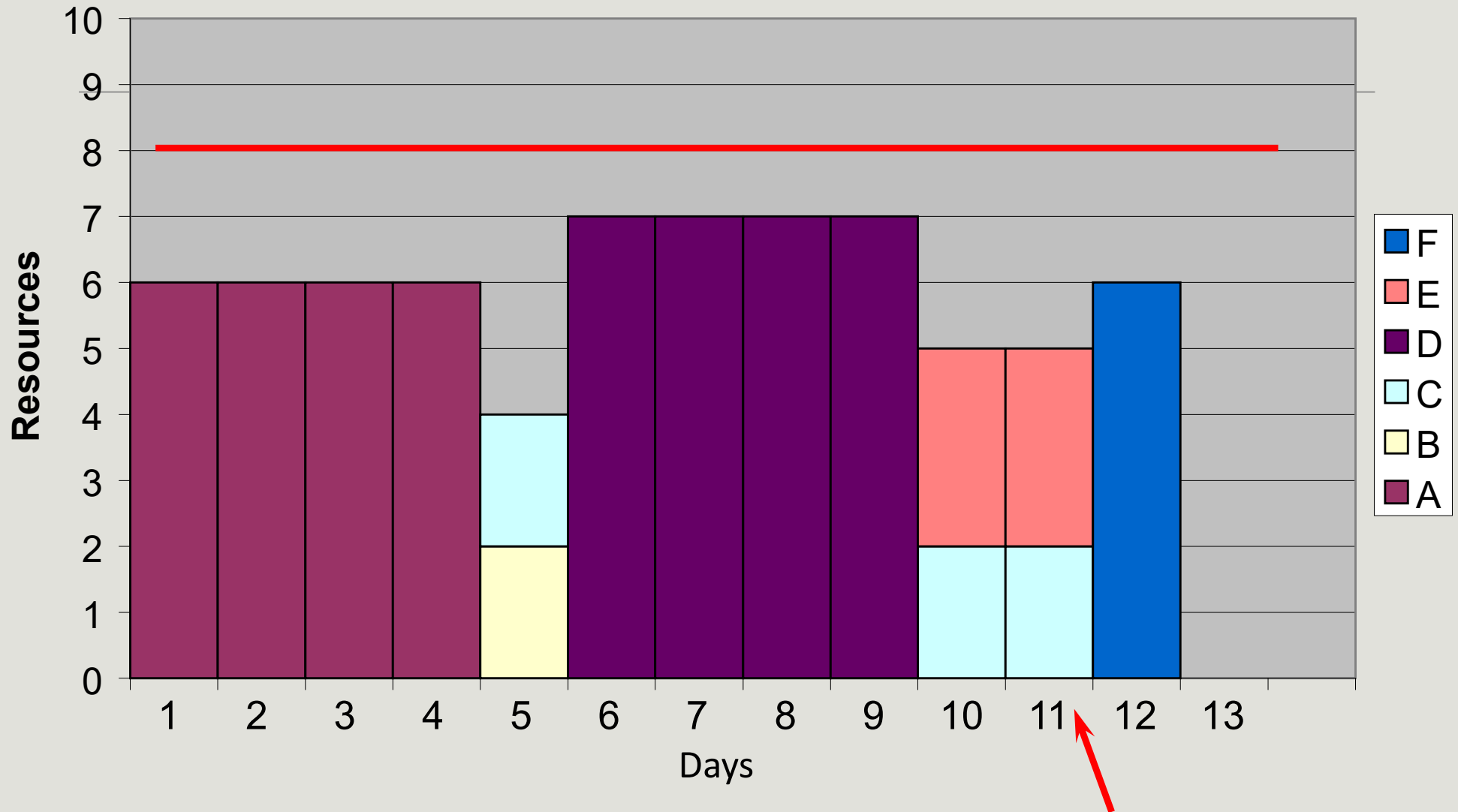
Resource Loading Charts

Resource loading charts used to display the amount of resources required as a function of time on a graph.

Each **activity** is typically represented in the charts as a block.



Resource Load Chart after Leveling and Splitting Task C



Splitting a task

Means **interrupting** the continuous stream of work on an activity at some point in its progress and applying that resource to another activity. Then returning to the original task.

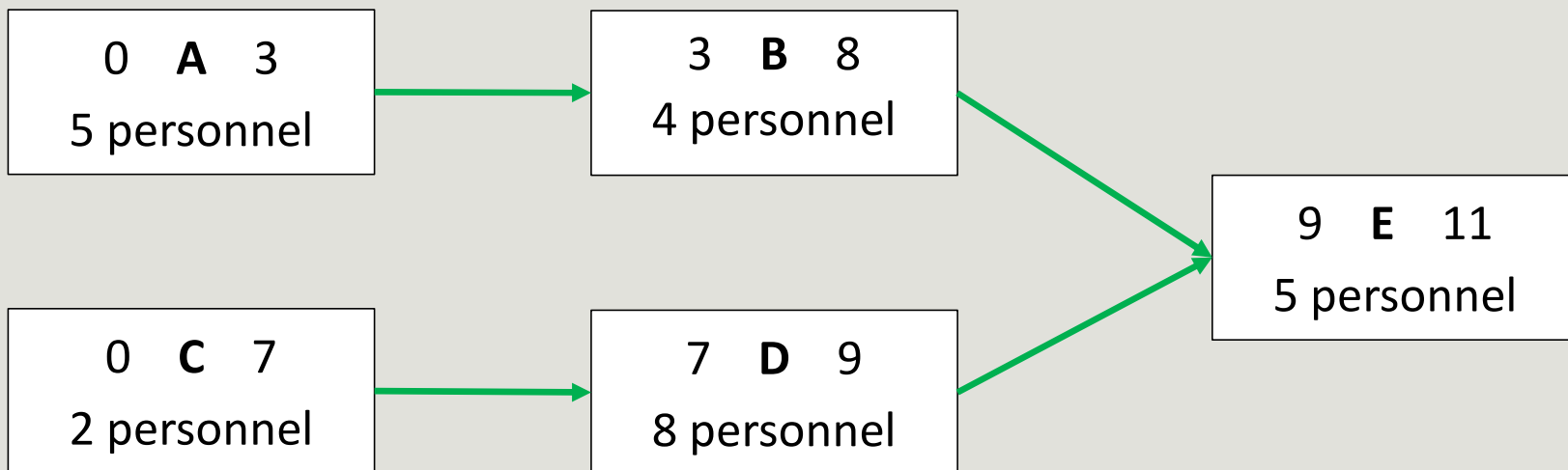
Splitting can be useful provided there are no excessive costs associated with splitting the task. For example, large **start-up** or **shutdown** costs make splitting an unattractive option.

Only some tasks can be split.

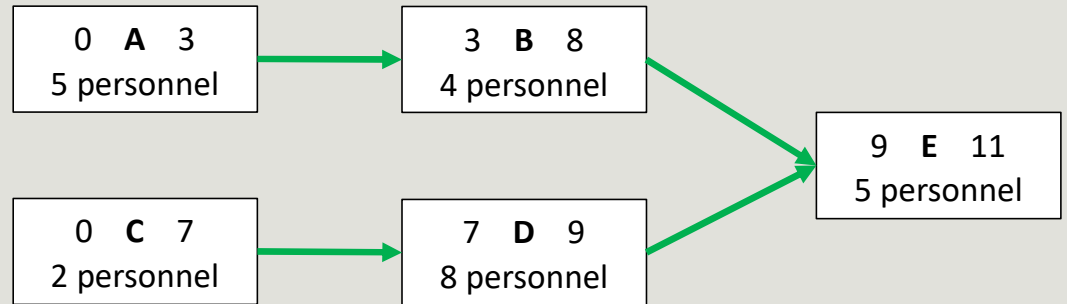


Resource Leveling Example

Create a Resource Load Chart from the following AON network diagram, and level it to 9 personnel per day. You can split the tasks.



Create a Resource Load Chart from the following AON network diagram, and level it to 9 personnel per day. You can split the tasks.



Resource Smoothing

Resource Smoothing is similar to **Resource Leveling**

Resource Leveling deals with a limited amount of a resource, a resource is only available at certain times, or the resource is over-allocated

In resource leveling it's sometimes necessary to extend the project duration (although this is not desired)

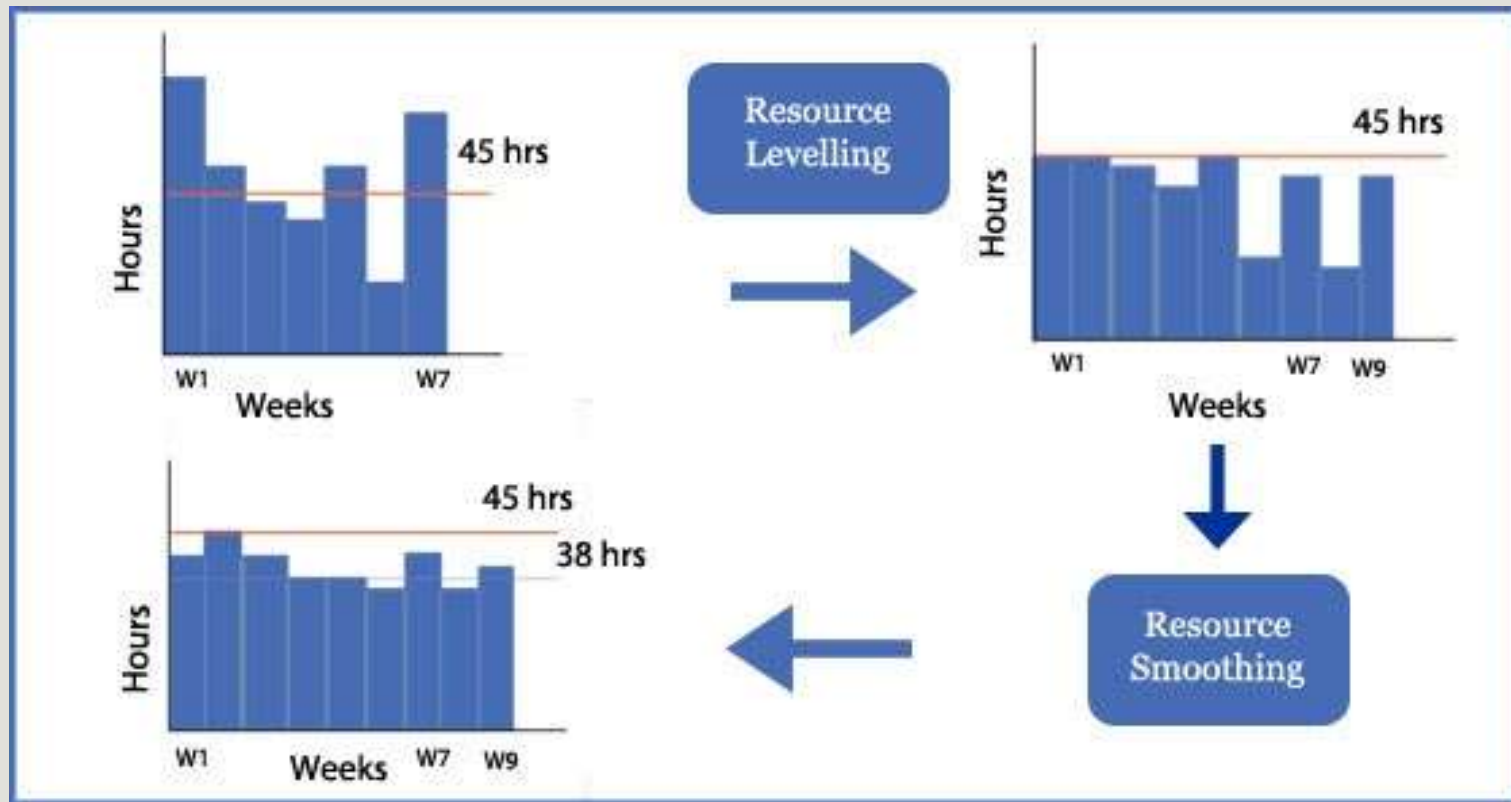
Resource Smoothing is done after Resource Leveling

It never extends the project duration, it just tries to use resources more efficiently by using available slack



Resource Smoothing

Here the project only has 45 hours of a resource per week, so it is **leveled** to that, but that extends the duration from seven weeks to nine weeks



<https://www.izenbridge.com>

After **resource smoothing**, the resource is used about the same each week (~38 hours)

Resource Allocation across Multiple Projects

Often different projects will use similar resources

Rather than acquire new resources for each project, your organization will usually **share resources** across **multiple projects**



<https://www.flickr.com/photos/vancouverfilmschool/5331017892>

Resource Allocation across multiple projects

The main challenge is **interdependency**:

- ❑ Any resource allocation decisions made in **one project** are likely to have ramifications in **other projects**.
- ❑ Some of the more common problems with interdependency among projects:
 - Inefficient use of resources;
 - Resource **bottlenecks**;
 - Ripple effects; and
 - Heightened pressure on personnel to multitask.



Resource Allocation across multiple projects

Any system used to optimize the complex problems with **multi-project resource allocation** must consider the need, and as much as possible, to minimize the negative effects of these three key factors:

1. **Schedule slippage**
2. **Resource Utilization**
3. **In-Process Inventory (*WIP*)**



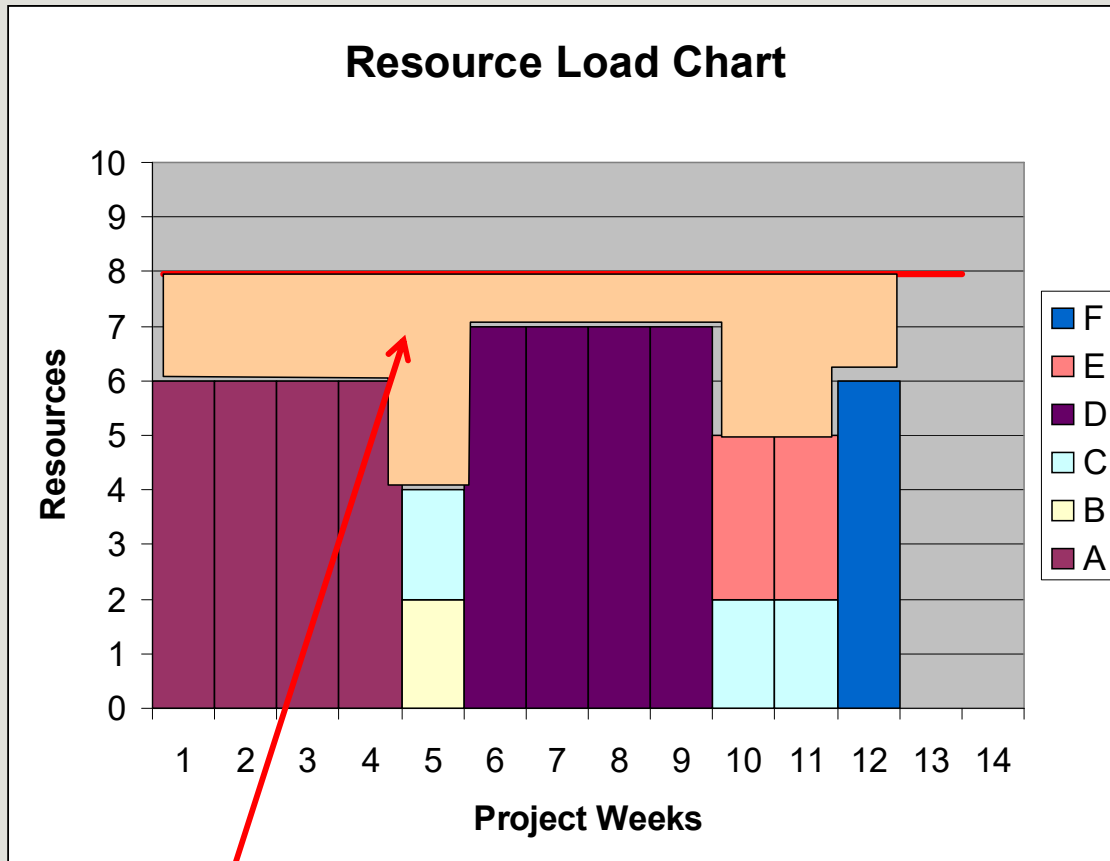
1. Schedule Slippage

For many projects this is more than the realization that the project will be late. In many industries this can result in **serious financial penalties**. It is not uncommon for companies to charge thousands of dollars in penalty clauses for each day a project is delayed.

In these cases, it is very important to consider the priority level a project gets based on the **impact of schedule slippage** for **each individual project**



2. Resource Utilization



Unused Resources

The goal of every company is to use the existing resources as efficiently as possible. Adding resources can be expensive and may not be necessary.

The **tan area** in the chart shows the **available resources** not used in the current project.

To maximize resource utilization, the available resources would be assigned to other, concurrent, projects thereby improving the overall efficiency with which we use project resources.

3. In-Process Inventory

The third standard for analyzing the **optimal use of multi project resources** is to consider their impact on **In-Process Inventory**. **In-Process Inventory** represents the amount of work waiting to be completed but **delayed due to unavailable resources**.

For example, an automotive company may find several resources delayed because it only has **one set of climate chambers for testing**. The projects stack up behind this resource bottleneck and represent the company's In-Process Inventory of projects.

Excessive In-Process Inventory is often caused by a lack of available resources and represents the kinds of trade-off decisions companies must make in multi project environments. Does it make sense to **hire more resources**? Is the bottleneck only **temporary**, leading to inefficient resource allocation later if you purchase new resources?



Resolving Resource Decisions in Multi-project Environments

There are several methods for resolving **resource allocation problems**:

"First in Line" or "first come-first served" is the simplest rule of thumb for allocating resources. It is easy to employ, it simply follows the master project calendar. The **earliest project** gets **scheduled first**. This method ignores any other important information such as priority, emergency situations, preference due to higher commercial success of other projects.

Greatest Resource Demand

With this decision rule the project with the **highest demand of resources** get **scheduled first**. Then the remaining projects are examined and scheduled according to their level of resource demand. The logic in this method is that in this way the **resource bottlenecks** will show up very quickly and it can be used as a starting point for potential additional resource allocation.



Resolving Resource Decisions in Multi-project Environments

Greatest Resource Utilization (*Most types of resources*)

Is a variation on the above rule. This rule schedules the project first that **uses the most types of resources**.

E.g. a company wants to run two projects A and B across a resource pool made up of programmers, system analysts and networking staff. Project A utilizes a lot of programmers. Project B utilizes all three resource types, but much less.

Using this rule, a company will prioritize project B above A in order to ensure that all resources are being utilized to the greatest extent.



Resolving Resource Decisions in Multi-project Environments

Minimum Late Finish Time (Earliest Finish First)

This rule prioritizes projects based on **finish dates**. The one with the **earliest late finish date** will be prioritized.



Resolving Resource Decisions in Multi-project Environments

Mathematical Programming

The common objectives these models seek to minimize are:

1. **Minimize** total development time for all projects;
2. **Minimize** resource utilization across all projects; and
3. **Minimize** total lateness across all projects.

While mathematical programming is a worthy approach to resolving the constrained resource problem, it tends to be limited due to the complexity of the problem, the large number of variables and the time necessary to generate a sufficiently small set of options.



Summary

Resource management in projects is a problem that is frequently overlooked by novice project managers or in firms that have not devoted enough time to understanding the full nature of the project management challenge they face.

As noted, it is common to develop project plans and schedules under the assumption of unlimited resources as if the organization can always find the trained personnel and other resources necessary to support project activities. This practice inevitably leads to schedule slippages and extra costs as the reality of resource availability overshadows the optimism of initial scheduling.

Use **Resource Management** to avoid these downfalls.



Discussion Questions

1. It has been argued that a project schedule which has **not been resource leveled** is useless. Do you agree or disagree with this statement? Why or why not?
2. How does resource smoothing affect multi-project scheduling?
3. Multi-tasking can have an important negative impact on your ability to resource level a project. When your team members are involved in multiple additional commitments, we must be careful not to assign their time too optimistically. In fact, it has been said: “**Remember, 40 hours is not the same as one week’s work.**” Comment on this idea. How does multi-tasking make it difficult to accurately resource level a project?
4. Why is resource management significantly **more difficult** in a multi-project environment? What are some rules of thumb to help project managers better control resources across several simultaneous projects?

Discussion Questions

5. Consider a project to build a **bridge** over a river gorge. What are some of the **resource constraints** that would make this project challenging?
6. Discuss the nature of “**time/cost tradeoffs**” on projects. What does this concept imply for our project management practices?
7. When resource leveling a project, there are several **heuristics** that can help us prioritize those activities that receive resources first. Explain how each of the heuristics works and give an example.
8. For many projects, the key resources to be managed are the **project team personnel**. Explain in what sense and how project team personnel are often the project critical resource.
9. What is the **philosophy** underlying resource loading? What does it do for our project? Why is it a critical element in effectively managing the project plan?